

# Laser Maze



## Concept:

Kids can assemble the components:

- **Goal:** no soldering
- **Easy concepts**
  - Send laser light
  - Reflect laser light
    - Multiple times
  - Sense laser light

- Close circuit
  - Sound alarm
- **Fun**
  - Different layouts
  - Exercise
  - Pretend you're the pink panther
  - Security for your secret hiding place
  - Keep your door open yet secure
- **Learnings**
  - Completing a circuit
  - Working with mirrors and reflections
  - What is a laser diode
  - What is a photo diode
  - What is a circuit
    - Electrical circuit
    - Light circuit (with mirrors)
- **Different Designs**
  - *Simple*: One mirror for simple door security
  - *Complex*: Perimeter with more mirrors
- **Maybe Next Class**
  - *Add more components*
  - *Get a text message when the beam is obstructed!*

## Cost:

under \$50 from Amazon

- Lasers (under \$.50 each)
- Mirrors
- Double stick tape
- Photo detector
- Battery
- Speaker
- Spray fog
- Miscellaneous

## Parts List:

- [Different lasers](#)
  - \$10 for a good laser
- [Batteries](#)
  - \$5
- [Speaker](#)
  - \$5

- [Mirrors](#)
  - \$6
- [Photo Diode](#)
  - \$7
- [Double Stick Tape](#)
  - \$10
- [Fog Spray](#)
  - \$20
- [Resistors](#)
  - \$10
- Miscellaneous

## Option: Repurposing Existing Kits

There are kits available for around \$50

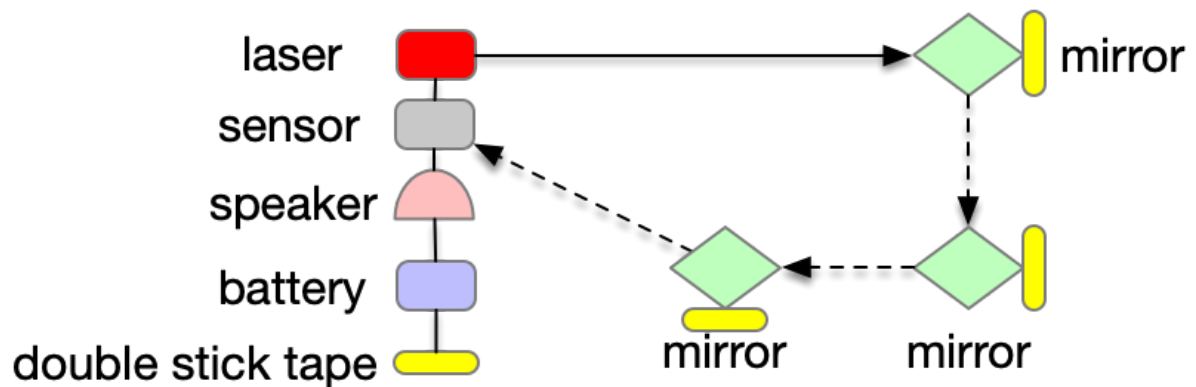
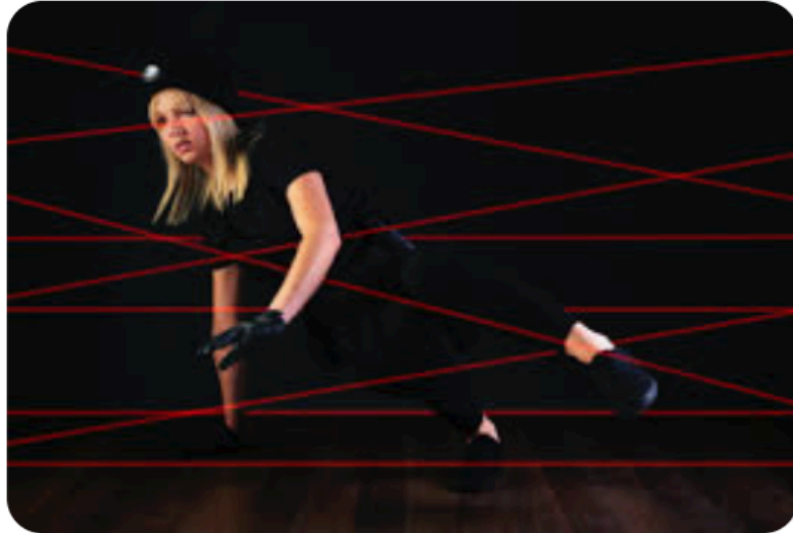
- [Existing Laser Security Kits for Kids](#)

## Shipping

- Everything can be ordered on Amazon and shipped directly to a third party (perfect)

Diagram

# Laser Maze



How to see the laser

[Diffusion Mist](#) or [Fog in a Can](#).



Amre Amer

## Notes:

If you like, You could put up two more simple projects. Like basic science fair like projects for middle school and high school kids with **sensors** detecting a b c like **fire** or pollution etc one in climate tech or projects like smart **plant** watering system or **obstacle** avoiding robot or drawing robot or **clap active light** or **traffic light** simulator or delivery robot. We have good demand in summer months.

Please let me know which hardware to purchase you will need a copy of the hardware I assume.